

A Salem Spectral Arc Answers Both Fourier-Dimension Questions in arXiv:2002.03855

Literature-implied full-result note
Run fa_banach_001, lane 6

11 August 2026

Status: literature-implied full answer. Li and Liu's later Salem spectral arc directly answers the second question. A short restriction-entropy computation, apparently not made in that paper, shows that the same measure also attains equality in Shi's necessary dimension inequality and therefore answers the first question.

The two source questions

At the end of Section 6 of Ruxi Shi, *On dimensions of frame spectral measures and their frame spectra*, arXiv:2002.03855 [1], the following questions are posed.

First, does there exist a singular continuous frame-spectral measure for which equality holds in

$$\sup_{\mu(K)>0, \mu(\partial K)=0} \dim_F \mu_K \leq \inf_{\mu(K)>0, \mu(\partial K)=0} \overline{\dim}_e \mu_K? \quad (1)$$

Second, does there exist a singular continuous frame-spectral measure with nonzero Fourier dimension?

Shi defines entropy dimension for probability measures. Thus, as is standard and as scale invariance requires, entropy dimension of a positive restriction means that of the normalized restriction $\mu_K/\mu(K)$. Fourier dimension is unchanged by multiplication by a positive scalar.

The later arc

Let $I = [-1/2, 1/2]$, let

$$\gamma(x) = (x, \sqrt{1-x^2}), \quad \sigma = \gamma_*(dx|_I),$$

so that

$$\int f d\sigma = \int_{-1/2}^{1/2} f(x, \sqrt{1-x^2}) dx.$$

Li and Liu exhibit exactly this measure in arXiv:2506.01280 [2]. They observe that it has orthonormal spectrum $\mathbb{Z} \times \{0\}$, and Section 10 proves

$$|\widehat{\sigma}(\xi)| \lesssim |\xi|^{-1/2}. \quad (2)$$

Indeed, the pullback of $e^{-2\pi i(n,0) \cdot y}$ under γ is $e^{-2\pi i n x}$, and these functions form the standard orthonormal basis of $L^2(I)$. Thus σ is spectral, hence frame-spectral.

The map γ is injective and Lipschitz on I . Consequently σ has no atoms, while its support is a planar arc of zero planar Lebesgue measure; hence it is singular continuous. The arc has Hausdorff dimension one. Equation (2) and the general inequality $\dim_F \sigma \leq \dim_H \text{supp } \sigma$ give

$$\dim_F \sigma = 1. \quad (3)$$

This already answers Shi's second question affirmatively.

Upgrade: all positive restrictions have entropy dimension one

Theorem. *For the measure σ above,*

$$\sup_{\sigma(K)>0, \sigma(\partial K)=0} \dim_F \sigma_K = 1 = \inf_{\sigma(K)>0, \sigma(\partial K)=0} \overline{\dim}_e \left(\frac{\sigma_K}{\sigma(K)} \right).$$

In particular equality holds in (1).

Proof. Fix a Borel set $K \subset \mathbb{R}^2$ with $\sigma(K) > 0$, and set

$$A = \gamma^{-1}(K) \subset I, \quad a = |A| > 0, \quad \nu = \frac{\sigma_K}{\sigma(K)}.$$

The first-coordinate marginal of ν is $a^{-1} \mathbf{1}_A(x) dx$. If J is a level- n dyadic interval, then its marginal mass p_J satisfies

$$p_J = \frac{|A \cap J|}{a} \leq \frac{2^{-n}}{a}.$$

For all sufficiently large n , Shannon entropy therefore satisfies

$$H((\pi_1)_* \nu, \mathcal{D}_n^{(1)}) = \sum_J p_J \log_2 \frac{1}{p_J} \geq \log_2 \frac{1}{\max_J p_J} \geq n + \log_2 a.$$

The first-coordinate partition is a coarsening of the planar dyadic partition, so

$$H(\nu, \mathcal{D}_n^{(2)}) \geq n + O(1). \quad (4)$$

Conversely, the graph of the Lipschitz function $x \mapsto \sqrt{1-x^2}$ on I meets at most $C2^n$ level- n dyadic squares. A probability vector supported on at most N atoms has entropy at most $\log_2 N$, whence

$$H(\nu, \mathcal{D}_n^{(2)}) \leq \log_2(C2^n) = n + O(1). \quad (5)$$

Equations (4)–(5) show that every positive restriction has entropy dimension exactly one. The null-boundary condition, needed in Shi's frame-restriction lemma, only narrows this class and does not alter the conclusion.

Every σ_K is supported on the same one-dimensional Lipschitz arc, so $\dim_F \sigma_K \leq 1$. On the other hand $K = \mathbb{R}^2$ is admissible, has empty boundary, and gives $\sigma_K = \sigma$; hence (3) makes the supremum equal to one. The entropy calculation makes the infimum equal to one. $\square$

Provenance and scope

The Salem spectral arc, its orthonormal spectrum, and the decay estimate are from Li–Liu. Their paper cites Shi for a different frame-density argument, but does not state that this arc answers Shi’s questions. The identification with the second question and especially the all-restrictions entropy calculation above are therefore recorded as a *literature-implied* answer, not as an author-explicit literature resolution and not as a new independent construction.

The conclusion is planar, exactly as Shi’s questions permit. It does not settle Li–Liu’s later, stricter real-line question.

References

- [1] R. Shi, *On dimensions of frame spectral measures and their frame spectra*, Ann. Fenn. Math. 46 (2021), 483–493; arXiv:2002.03855. <https://arxiv.org/abs/2002.03855>.
- [2] L. Li and B. Liu, *Fourier Frames on Salem Measures*, arXiv:2506.01280 (2025), especially the introduction and Section 10. <https://arxiv.org/abs/2506.01280>.